

CHACALLUTA, Chile

WMO: 854060

Lat: 18.3514S

Lon: 70.3358W

Elev: 180

StdP: 14.60

Time Zone: -4.00 (CHL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	53.6	55.5	46.0	46.2	55.6	46.9	47.9	56.9	18.0	65.1	16.3	64.1	5.7	80	0.328

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.6	80.3	69.2	78.8	67.8	77.4	67.1	71.5	78.0	69.7	76.7	68.3	75.7	12.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.6	105.7	76.8	66.5	98.4	75.0	64.8	92.6	73.8	35.3	77.8	33.9	76.5	32.6	76.1	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	16.4	15.0	DB	49.5	83.3	2.4	2.3	47.8	85.0	46.4	86.3	45.0	87.6	43.3	89.3	(4)
(5)			WB	46.5	70.9	2.0	2.1	45.0	72.4	43.9	73.6	42.8	74.8	41.3	76.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	66.6	72.8	73.4	71.8	68.1	65.0	62.6	61.0	61.2	62.3	64.4	67.0	69.9	(6)
(7)		DBStd	4.91	2.31	2.15	2.28	2.55	2.85	2.36	2.41	2.18	2.11	2.11	2.00	2.19	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	492	0	0	0	4	36	78	125	121	86	36	6	0	(9)
(10)		CDD50	6055	706	656	676	544	464	377	342	348	369	446	510	617	(10)
(11)		CDD65	1070	241	236	211	98	34	5	2	3	4	18	66	152	(11)
(12)		CDH74	1985	627	713	459	66	9	0	0	3	0	0	2	106	(12)
(13)		CDH80	44	14	20	8	0	1	0	0	1	0	0	0	0	(13)
(14)	Wind	WSAvg	6.3	6.9	6.6	6.4	6.0	5.7	5.4	5.3	5.8	6.5	7.0	7.2	7.0	(14)
(15)	Precipitation	PrecAvg	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(15)
(16)		PrecMax	0.8	0.3	0.1	0.1	0.0	0.1	0.1	0.4	0.2	0.3	0.1	0.0	0.1	(16)
(17)		PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.1	0.0	0.0	0.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.0	82.3	80.9	77.3	75.1	71.2	69.4	70.1	70.1	71.8	73.8	77.4	(19)
(20)			MCWB	71.5	70.6	69.3	66.7	66.0	62.8	61.6	63.8	63.3	63.4	63.6	66.7	(20)
(21)		2%	DB	79.2	80.5	79.0	75.6	72.6	68.4	66.5	66.6	67.9	70.0	73.1	76.3	(21)
(22)			MCWB	68.7	68.9	68.1	65.5	63.8	60.6	59.9	60.7	61.3	61.8	63.2	66.0	(22)
(23)		5%	DB	78.5	78.9	77.4	74.6	71.3	67.6	65.8	64.8	66.3	68.4	71.8	75.2	(23)
(24)			MCWB	67.9	67.6	67.0	64.8	62.9	60.1	59.3	58.1	59.9	60.7	62.3	65.0	(24)
(25)		10%	DB	77.1	77.4	76.7	73.2	69.8	66.3	64.4	64.1	65.0	67.9	70.8	73.8	(25)
(26)			MCWB	67.2	67.1	66.6	64.0	62.0	59.3	57.9	57.4	58.5	60.4	61.9	64.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.5	72.6	71.5	68.2	67.2	64.2	62.7	65.0	64.4	65.6	65.6	69.7	(27)
(28)			MCDWB	79.1	79.0	78.2	75.2	73.0	68.3	66.9	69.6	68.1	68.8	71.3	75.0	(28)
(29)		2%	WB	72.0	71.0	69.6	66.8	64.9	61.9	61.1	61.7	62.5	63.6	64.5	67.8	(29)
(30)			MCDWB	78.1	78.0	76.8	74.1	71.4	67.2	65.8	66.5	66.8	67.7	70.9	74.0	(30)
(31)		5%	WB	70.0	69.4	68.4	65.5	63.5	60.6	59.8	59.2	60.7	62.1	63.5	66.3	(31)
(32)			MCDWB	76.6	76.9	75.8	72.9	69.7	65.9	64.6	63.4	65.4	67.1	70.1	73.5	(32)
(33)		10%	WB	67.9	68.1	67.3	64.6	62.4	59.8	58.5	57.9	59.2	60.8	62.6	65.2	(33)
(34)			MCDWB	75.3	76.1	75.1	71.9	68.6	65.3	63.3	62.5	64.1	66.9	69.4	72.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.3	9.6	10.1	10.1	9.2	7.0	6.1	5.5	5.9	6.7	7.9	8.9	(35)
(36)			MCDBR	9.4	9.7	10.3	10.9	10.5	8.4	6.8	6.1	6.6	7.2	8.4	8.8	(36)
(37)			MCWBR	4.9	5.0	5.5	6.4	6.3	5.2	4.2	3.6	3.7	4.0	4.7	4.9	(37)
(38)		5% WB	MCDWB	7.9	8.7	9.7	10.0	10.2	8.0	6.3	6.0	6.4	7.1	8.1	8.4	(38)
(39)			MCWBR	4.3	4.9	5.6	6.0	6.7	5.3	4.5	3.9	4.0	4.4	4.9	5.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.515	0.521	0.505	0.452	0.404	0.385	0.386	0.400	0.417	0.428	0.432	0.483	(40)	
(41)		taud	2.069	2.055	2.094	2.199	2.274	2.327	2.323	2.267	2.219	2.201	2.218	2.110	(41)	
(42)		Ebn at Noon	267	262	259	261	262	262	266	272	279	285	288	276	(42)	
(43)		Edh at Noon	56	56	53	44	38	35	36	41	46	48	48	54	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2388	2279	2188	1835	1468	1191	1192	1443	1775	2067	2388	2478	(44)	
(45)		RadStd	133	156	108	143	77	81	84	97	114	92	106	111	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page